Name:-Keya Jadhav
PRN:-123B1B142

CODE:-

```
#include <iostream>
#include <string>
using namespace std;
class Vehicle
{ protected:
string make, model; int year;
public:
void get_vehicle_data()
 {
 cout << "Enter make and model of the vehicle: " << endl;
cin >> make >> model;
 cout << "Enter year of manufacture: ";
cin >> year;
 }
 void display_data()
 {
 cout << "make: " << make << endl;
cout << "model: " << model << endl;
 cout << "year of manufacture: " << year << endl;
 }
};
class Car : public Vehicle
{ private:
 int seating_capacity, warranty = 4;
string fuel_type;
public: void get_car_data()
 {
 get_vehicle_data();
 cout << "Enter seating capacity of car: ";
cin >> seating_capacity;
cout << "Enter fuel type of car: ";
 cin >> fuel_type;
 }
 void display_car_data()
 {
 display_data();
 cout << "Seating capacity: " << seating_capacity << endl;
cout << "Warranty provided is " << warranty << " years" << endl;
 cout << "Fuel type: " << fuel_type << endl;
 }
};
class Motor_Cycle : public Vehicle
{ private:
string wheelType, coolingType;
public: void get_MotorCycle_data()
 {
 get_vehicle_data();
```

```
cout << "Enter wheel type and cooling type of motorcycle:" << endl;
 cin >> wheelType >> coolingType;
 }
 void display_MotorCycle_data()
 {
 display_data();
 cout << "Wheel type: " << wheelType << endl;
 cout << "Cooling type: " << coolingType << endl;
 }
}; int main()
{
 Car c;
 Motor_Cycle m;

 c.get_car_data();
cout << "CAR DETAILS" << endl;
 c.display_car_data();
 cout << endl;
 m.get_MotorCycle_data();
cout << "MOTORCYCLE DETAILS" << endl;
 m.display_MotorCycle_data();
 return 0;
}
```

Name:-Keya Jadhav
PRN:-123B1B142

OUTPUT:-

```
Enter make and model of the vehicle:
BMW
C18
Enter year of manufacture: 2022
Enter seating capacity of car: 4
Enter fuel type of car: Petrol
CAR DETAILS
make: BMW
model: C18
year of manufacture: 2022
Seating capacity: 4
Warranty provided is 4 years
Fuel type: Petrol

Enter make and model of the vehicle:
Bugatti
R-35
Enter year of manufacture: 2022
Enter wheel type and cooling type of motorcycle:
MRF
Air
MOTORCYCLE DETAILS
```

make: Bugatti
model: R-35
year of manufacture: 2022
Wheel type: MRF
Cooling type: Air
Enter make and model of the vehicle:
Xeon G-20
Enter year of manufacture: 2023
Enter seating capacity of car: 5
Enter fuel type of car: EV
CAR DETAILS
make: Xeon
model: G-20
year of manufacture: 2023
Seating capacity: 5
Warranty provided is 4 years
Fuel type: EV

Enter make and model of the vehicle:
Ducati X-5
Enter year of manufacture: 2021
Enter wheel type and cooling type of motorcycle:
MRF Oil
MOTORCYCLE DETAILS
make: Ducati
model: X-5
year of manufacture: 2021
Wheel type: MRF
Cooling type: Oil